# Business and Finance

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All businesses should produce some amount of *revenue*: the total amount of income generated by the sale of goods and services related to the primary operations of the business. Revenue taken in is like a big pool of cash that the company can do things with. One key part is paying for the *cost of goods sold* (COGS): the direct costs associated with producing the goods and services sold. For example, if you run a lemonade stand, what is the cost of the lemons, sugars, cups, etc. that you need in order to sell your product. Or, if you are a database company, this might include the cost to run servers or the underlying infrastructure that supports your service. This is money you must pay in order to function and sell a product.

## Gross Margin

A core related concept is *gross margin*: the difference between total revenue and the cost of goods sold. If you take in $100 in revenue and your COGS is $10, then your gross margin is $90. Typically, it makes sense to present this proportionally, as a percentage of revenue. That is,

$$\text{Gross Margin} = \frac{Revenue - COGS}{Revenue} \times 100 \tag{1}$$

Money left over after paying for COGS is called *gross profit*. This is then used to go ahead and pay for other expenses, like labor, sales/marketing, etc.

If you then take the *gross profit* and deduct these other expenses, like R&D, sales/marking, etc. then this gives you the *operating margin* i.e telling you how much is left over after cost of goods sold and cost to basically run the business.

$$\text{Operating Margin} = \frac{GrossProfit - OperatingExpenses}{Revenue} \times 100 \tag{2}$$

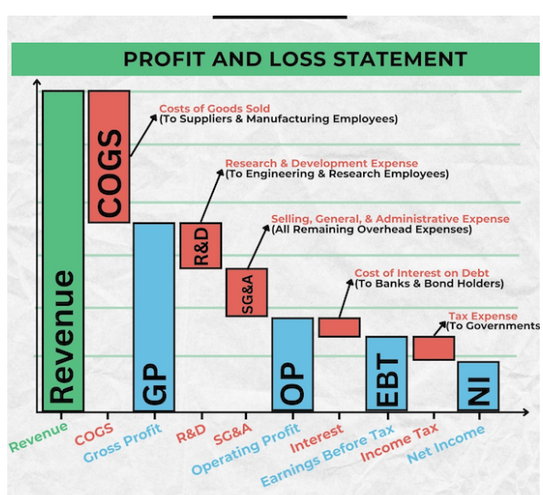


Figure 1: Gross margin and operating margin illustrated as an overall profit and loss statement.

## References